FluxDone

Product Requirements Document (PRD)

Version 1.0 | March 2026 | Flux App Family

1. Project Overview

App Name	FluxDone
Abbreviation	FD
App Family	Flux (FluxDone + FluxFocus)
Platform	Android (Phase 1) → iOS + Web (Phase 2)
Framework	Flutter (Dart)
Storage	Local SQLite / Hive (offline-first)
Sync	Google Drive DB backup (post-offline phase)
Version	1.0 — PRD
Author	Ari
Document Type	Product Requirements Document

FluxDone is a fully offline, highly customizable personal task management application built as a direct replacement for TickTick. It is designed around the specific academic workflow of a student preparing simultaneously for Math Olympiad (IOQM → INMO → IMO), JEE Foundation (Neev Diamond batch), ICSE school curriculum, and programming. FluxDone is the first of two applications in the Flux family; it communicates with FluxFocus (the focus and screen discipline app) to enable automatic focus block scheduling when tasks are created.

The application is built with an offline-first philosophy — all core functionality operates without any network connection. Cloud features (Google Drive backup, Google Calendar read-only sync) are supplementary and non-blocking. The architecture is designed to be strictly modular so that each feature can be developed, tested, maintained, and broken independently without affecting the rest of the application.

2. Product Vision

2.1 Vision Statement

FluxDone exists to give the user complete ownership of their task and time management system — without subscriptions, advertisements, forced updates, or features they did not ask for. Every interaction is intentional, every color is meaningful, and every workflow maps directly to how the user actually thinks about their work.

2.2 Design Philosophy

- Offline-first: The app must be fully functional with zero network connectivity. Network features are enhancements, not dependencies.
- Modular architecture: Every feature is an independent module. A failure or change in one module does not cascade to others.
- Clone-first, improve-later: Phase 1 replicates the proven TickTick workflow exactly. Improvements and new patterns are introduced in Phase 2 only after the user validates the base experience.
- Visual identity through color: The color system is not decorative — it is a cognitive tool. Colors map directly to academic domains and are mirrored in companion app Notewise.
- 70–80% feature density: The UI is closer to feature-rich than minimal, but every visible element earns its place.
- No feature bloat: Features not used by the user are not built. Pomodoro, attachments, collaboration, and filters are explicitly excluded from scope.

2.3 Success Definition

FluxDone is successful when the user can fully migrate away from TickTick and experience no loss of productivity, combined with a gain in workflow ownership and visual customization control.

3. Target User

3.1 Primary User Profile

Name	Ari
Academic Stage	ICSE Class 10 → 12
Olympiad Track	IOQM → INMO → IMO
Coaching	Neev Diamond batch (PW) — JEE Foundation (Physics, Chemistry, Math)
Device	Android smartphone (primary), tablet, laptop (future sync targets)
App Mode	Mobile only, light mode (Phase 1), dark mode (Phase 2)

Skill Level	Python beginner; directs AI-generated code; understands architecture and logic

3.2 User Behavioral Patterns

- Creates the majority of timed tasks directly inside the Calendar view by tapping empty time slots on the timeline.
- Uses calendar in single-day, 3-day, and weekly formats interchangeably depending on planning horizon.
- Organizes all academic domains by color — Number Theory (deep green), Geometry (royal blue), Combinatorics (fresh green), Algebra P1 (orange), Algebra P2 (red-orange), Tests (red), Neev Diamond (slate-indigo).
- Writes detailed rich-text descriptions for most non-recurring tasks and brief descriptions for approximately half of recurring tasks.
- Tags difficulty on physical books and PDFs, not inside the task app — FluxDone does not need to handle difficulty tracking.
- Uses Google Calendar (not local device calendar) for birthdays, school events, and niche personal tasks — needs these visible in FluxDone without duplication.
- Uses Smart Lists (Today, Tomorrow, Upcoming, All, Completed, Trash) daily for task triage.
- Uses recurring tasks frequently — on par with Habit Tracker usage.

4. Core Features

4.1 Feature Priority Legend

Priority	Definition
P0	Must ship in Phase 1. App is non-functional without this.
P1	Ships in Phase 1. Significantly degrades experience if absent.
P2	Ships in Phase 2. Enhancement or quality-of-life improvement.

4.2 Feature Table

ID	Feature	Description	Priority
FD-01	**Task Creation**	Create a task with a mandatory title, mandatory date, optional start time, optional end time (duration), optional reminder, and optional recurrence rule. Tasks	**P0**

ID	Feature	Description	Priority
		with a start and end time are rendered as visual blocks on the timeline in Calendar View.	
FD-02	**Task Completion**	Mark any task as complete with a single tap on its checkbox. Completed tasks move to the Completed smart list and are visually struck through.	P0
FD-03	**Task Deletion & Trash**	Deleting a task moves it to Trash (soft delete). Trash retains tasks for 30 days. User can permanently delete from Trash or restore to original list.	P0
FD-04	**Rich Text Description**	Every task has an optional description field supporting markdown-style formatting: headings (H1, H2, H3), unordered lists, ordered lists, bold, italic, and inline code. Formatting is triggered by symbol combinations (e.g., ## for H2, ** for bold) and renders inline as the user types.	P0
FD-05	**Subtasks**	A task can contain an ordered list of subtasks. Each subtask has a title and a completion checkbox. Subtasks do not have their own due dates or reminders. Parent task completion does not auto-complete subtasks.	P1
FD-06	**Priority Flags**	Each task can be assigned one of four priority levels: None, Low, Medium, High. Priority is displayed as a colored flag icon on the task card. Tasks can be sorted by priority within a list.	P1
FD-07	**Recurring Tasks**	Tasks can recur on a custom schedule: daily, weekly (specific days), monthly (specific date or day-of-week), or a custom interval in days. Each recurrence generates a new task instance. Editing a recurring instance can apply the change to: this instance only, this and future instances, or all instances.	P0
FD-08	**Reminders**	Each task supports one or more reminders set as absolute times or relative offsets before the task start time (e.g., 15 minutes before, 1 hour before). Reminders fire as local device push notifications and do not require network connectivity.	P1
FD-09	**Folders**	The top level of organization. A Folder contains one or more Lists. Folders can be expanded or collapsed in the sidebar. Each Folder has a display name.	P0
FD-10	**Lists**	A List belongs to exactly one Folder. A List contains tasks and optional Sections. Each List has a display name and a custom color (full hex color picker). Lists are the primary organizational unit.	P0
FD-11	**Sections**	A Section is an optional named divider within a List that groups tasks. Sections are used to subdivide a list by subject (e.g., Physics, Chemistry, Math inside	P1

ID	Feature	Description	Priority
		the Neev Diamond list). Tasks can be moved between Sections via drag-and-drop or context menu.	
FD-12	**Custom Colors**	Every List has a fully customizable color via a hex color picker. Colors are displayed as a circular swatch next to the list name in the sidebar, as a colored left-border accent on task cards in list view, and as the fill color of task blocks on the calendar timeline.	P0
FD-13	**Smart Lists**	System-generated read-only views: Today (tasks with date = today), Tomorrow (tasks with date = tomorrow), Upcoming (tasks with date in the next 7 days), All (all non-completed, non-trashed tasks), Completed (all completed tasks), Trash (soft-deleted tasks). Smart Lists are listed in the sidebar below user lists.	P0
FD-14	**Calendar View — Timeline**	A vertical scrollable timeline showing tasks as colored duration blocks. Supports three view modes: single-day, 3-day, and weekly. Tapping an empty time slot opens the task creation sheet with start time pre-populated. Overlapping tasks are displayed side by side (column layout), identical to Google Calendar's overlap rendering.	P0
FD-15	**Calendar View — Task Creation**	Tasks created inside Calendar View have their start time and date pre-filled based on the tapped slot. The user can drag the bottom edge of a new block to set the end time (duration) before saving.	P0
FD-16	**Calendar View — Google Calendar Overlay**	Read-only events from the user's linked Google Calendar account are displayed on the timeline as non-interactive blocks with a distinct visual style (e.g., outlined border, muted color) to differentiate them from FluxDone tasks. Google Calendar events cannot be edited from within FluxDone.	P1
FD-17	**Habit Tracker**	A separate screen for tracking recurring habits. Each habit has: a display name, a custom color, a frequency rule (daily, specific days of week, X times per week), a streak counter, a completion history (calendar heatmap or grid), and a daily completion toggle. Habits are independent from the task system.	P1
FD-18	**Navigation — Hamburger Drawer**	A hamburger button in the top-left corner of every screen opens the side drawer. The drawer contains: all Folders and their nested Lists, all Smart Lists, and a section for app settings. Tapping a List navigates to its task view. The drawer slides in from the left and dismisses on outside tap or back gesture.	P0
FD-19	**Navigation — Bottom Navigation Bar**	In portrait orientation, a bottom navigation bar provides access to: List/Task View, Calendar View, Habit Tracker, and Settings. In landscape orientation, this bar migrates to the left edge of the screen as a vertical icon rail (Windows/Linux taskbar behavior).	P0

ID	Feature	Description	Priority
FD-20	**Light and Dark Theme**	The app supports both light and dark themes. The user can toggle between them in Settings. Phase 1 ships with light mode as default. Dark mode is added in Phase 2.	P1
FD-21	**Google Drive Backup**	The user can link their Google account. FluxDone creates a dedicated folder in their Google Drive and uploads a snapshot of the local database on demand or on a scheduled interval. Any device with the same Google account can download and restore the database, enabling cross-device sync. This feature is built after all offline features are complete.	P2
FD-22	**FluxFocus Integration**	When creating a task with a defined start time and end time, the user is optionally prompted to create a matching focus block in FluxFocus. Inter-app communication uses Android's Intent system or a shared local IPC mechanism. This is specced in the inter-app communication document.	P2

5. Screen Inventory

5.1 Screen List

Screen ID	Screen Name	Purpose
SCR-01	**List View**	Displays all tasks in the currently selected List or Smart List. Tasks are grouped by Section if sections exist. Each task card shows title, priority flag, date, and a colored left-border accent matching the list color.
SCR-02	**Task Detail / Edit Sheet**	Bottom sheet or full-screen view showing all task fields: title, date, start time, end time, reminder(s), recurrence, priority, subtasks, and rich-text description. Opened by tapping any task card.
SCR-03	**Task Creation Sheet**	Same layout as Task Detail but in creation mode. Pre-populated with context data when opened from Calendar View (start time, date).
SCR-04	**Calendar View**	Full-screen interactive timeline. Includes view mode toggle (day / 3-day / week), current date navigation arrows, and the timeline canvas with task blocks and Google Calendar overlay blocks.
SCR-05	**Habit Tracker Screen**	Displays all active habits with their streak, today's completion status, and a toggle. Includes a button to add a new habit and access to habit history.

Screen ID	Screen Name	Purpose
SCR-06	**Habit Detail / Edit**	Full detail view for a single habit: name, color, frequency, streak, and full history heatmap.
SCR-07	**Habit Creation Sheet**	Sheet for creating a new habit: name, color, frequency rule.
SCR-08	**Side Drawer**	Slides in from the left. Contains: app logo/name, all Folders with nested Lists, Smart Lists section, and Settings shortcut.
SCR-09	**Settings Screen**	App-level settings: theme toggle (light/dark), Google Account link, Google Calendar connection, Google Drive backup controls, notification permissions, app version info.
SCR-10	**Folder Management**	Create, rename, reorder, and delete Folders. Accessible from the side drawer via long-press or edit icon.
SCR-11	**List Management**	Create, rename, change color, reorder, and delete Lists. Accessible from the side drawer.
SCR-12	**Trash Screen**	Displays all soft-deleted tasks. Options to restore individual tasks or permanently delete. Bulk empty-trash action.
SCR-13	**Google Drive Backup Screen**	Shows last backup timestamp, device-linked Google account, manual backup trigger, and restore-from-backup option.

6. Key User Flows

6.1 Create a Timed Task from Calendar View

1. User opens FluxDone. The bottom navigation bar is visible.
2. User taps the Calendar icon in the bottom navigation bar. SCR-04 (Calendar View) opens.
3. User selects the desired view mode (single-day, 3-day, or weekly) using the toggle at the top of the screen.
4. User visually scans the timeline to identify an empty time slot that does not conflict with existing task blocks or Google Calendar overlay blocks.
5. User taps the desired empty time slot on the timeline. SCR-03 (Task Creation Sheet) opens with the date and start time pre-populated from the tapped position.
6. User drags the bottom edge of the new task block preview on the timeline to set the desired end time. Alternatively, the user manually enters the end time in the creation sheet.
7. User enters the task title. Optionally adds description, priority flag, reminder, subtasks, and recurrence.
8. User taps Save. The task is persisted to local storage and appears immediately as a colored block on the timeline. The color matches the List to which the task was assigned.

9. If the task has a defined start and end time, the user is optionally prompted: “Create a matching focus block in FluxFocus?” The user can accept or dismiss.

6.2 Create a Task from List View

10. User opens the hamburger drawer and taps a List name (e.g., Number Theory). SCR-01 (List View) opens showing that list’s tasks.
11. User taps the floating action button (FAB) or the inline “+ Add Task” row at the bottom of the list. SCR-03 (Task Creation Sheet) opens.
12. User fills in title, date, and any optional fields. Saves.
13. Task appears in the list under the appropriate Section (if sections exist) or at the bottom of the list.

6.3 Daily Planning Flow

14. User opens the Today smart list from the side drawer. SCR-01 shows all tasks due today across all lists, color-coded by their source list.
15. User reviews the list and switches to Calendar View to see the timeline layout of today’s tasks.
16. User identifies gaps or conflicts and creates or reschedules tasks directly on the timeline.
17. User checks off completed tasks throughout the day. Completed tasks move to the Completed smart list.

6.4 Habit Check-in Flow

18. User taps the Habit Tracker icon in the bottom navigation bar. SCR-05 opens.
19. User sees all habits listed with today’s completion toggle and current streak.
20. User taps the toggle for each completed habit. The streak counter increments.
21. User can tap any habit to open SCR-06 (Habit Detail) and view the full completion history heatmap.

7. Success Metrics

- User successfully completes full migration from TickTick with zero loss of existing task data.
- All P0 features functional and stable on Android before any P1 or P2 work begins.
- Calendar timeline renders, allows task creation, and handles overlapping tasks correctly on all supported Android screen sizes.
- Rich-text description editor correctly renders all supported formatting types (H1–H3, bold, italic, lists).
- Local notifications (reminders) fire correctly without network connectivity.

- Google Calendar overlay loads and displays read-only events without blocking app launch.
- App performs all core operations (task creation, list navigation, calendar rendering) in under 200ms on a mid-range Android device.
- Google Drive backup successfully uploads and restores the full local database without data corruption.
- FluxDone and FluxFocus communicate successfully via the inter-app IPC mechanism on the same device.

8. Out of Scope

The following features are explicitly excluded from FluxDone and will not be built in any phase:

- Pomodoro / built-in focus timer (handled by FluxFocus)
- File or image attachments on tasks
- Tags or tag-based filtering
- Task collaboration or sharing with other users
- Advanced filter builder (filter by priority, date range, list, etc.)
- Natural language task parsing (e.g., “Meet tomorrow at 3pm” auto-filling fields)
- AI-powered suggestions or smart scheduling
- Widget support (home screen or lock screen) — Phase 1 only
- Apple Calendar or local device calendar sync (Google Calendar only)
- Task templates
- In-app browser or URL preview

9. Development Phases

Phase 1 — Offline Core (Android)

All P0 features must be complete and stable before Phase 1 is considered done. P1 features are built after P0 is validated.

- FD-01: Task Creation
- FD-02: Task Completion
- FD-03: Task Deletion & Trash
- FD-04: Rich Text Description
- FD-07: Recurring Tasks
- FD-09: Folders
- FD-10: Lists
- FD-12: Custom Colors
- FD-13: Smart Lists
- FD-14: Calendar View — Timeline

- FD-15: Calendar View — Task Creation
- FD-18: Hamburger Drawer Navigation
- FD-19: Bottom / Left Navigation Bar

P1 features added after P0 validation:

- FD-05: Subtasks
- FD-06: Priority Flags
- FD-08: Reminders
- FD-11: Sections
- FD-16: Google Calendar Overlay
- FD-17: Habit Tracker
- FD-20: Dark Theme

Phase 2 — Sync & Integration

- FD-21: Google Drive Backup & Cross-Device Sync
- FD-22: FluxFocus Inter-App Integration
- iOS platform support
- Web platform support

10. Privacy & Safety

- All task and habit data is stored exclusively in local device storage (SQLite/Hive). No data is sent to any server in Phase 1.
- Google Calendar access is read-only. FluxDone never writes to, modifies, or deletes Google Calendar events.
- Google Drive backup in Phase 2 uploads only the local database file. No metadata, analytics, or usage data is transmitted.
- Google account authentication in Phase 2 uses OAuth 2.0 via the official Google Sign-In SDK. No credentials are stored by the app.
- The app requests only the minimum Android permissions required: local notifications, storage access (for DB backup), and internet (for Google features only).
- No analytics, crash reporting, or telemetry is collected in Phase 1.

11. Definition of Done

A feature is considered “Done” when all of the following criteria are met:

22. The feature works correctly on an Android physical device or emulator (API 26+).
23. The feature is implemented as an isolated module with no tight coupling to unrelated modules.

24. The feature works correctly with zero network connectivity (for all non-network features).
25. Edge cases are handled: empty states, long text, zero tasks, maximum tasks, back navigation, and screen rotation.
26. The feature has been reviewed and approved by the user (Ari) in a working build.
27. No regression has been introduced in any previously approved feature.

12. Design System

12.1 Academic Domain Color Map

Domain / List	Color Name	Hex Code	Notes
Number Theory	Deep Green	`#2E7D32`	
Geometry	Royal Blue	`#1565C0`	
Combinatorics	Fresh Green	`#43A047`	*Probability grouped here*
Algebra P1	Orange	`#FB8C00`	*Basic Math, Polynomials, Quadratics, FE, Binomial*
Algebra P2	Red-Orange	`#E64A19`	*Sequences, Recurrences, Inequalities*
Tests	Red	`#E53935`	*Mock tests and exam practice*
Neev Diamond	Slate-Indigo	`#576481`	*JEE Foundation — Physics, Chemistry, Math*
Programming	TBD	`TBD`	*Must contrast with Neev Diamond*
Short Notes	TBD	`TBD`	*High visibility recommended*
School	TBD	`TBD`	
Doubts	TBD	`TBD`	

12.2 Color Design Principles

- Number Theory and Combinatorics share the green family to reflect their combinatorial nature.
- Geometry stands alone in blue — visually distinct, spatially associated.
- Algebra uses the warm spectrum (red → orange → yellow family) to create a gradient between related but distinct sub-domains.

- Neev Diamond (JEE batch) uses a neutral slate-indigo to remain distinct from all Olympiad domain colors.
- All colors are validated for visibility in light mode. Dark mode validation occurs in Phase 2.
- Colors are mirrored exactly in Notewise folders to enable cross-app cognitive association.

12.3 Typography

- Primary font: System default (Roboto on Android).
- Task titles: 16sp, regular weight.
- Section headers: 14sp, medium weight, muted color.
- Description body: 14sp, regular weight.
- Metadata (date, time, priority): 12sp, regular weight, secondary color.

12.4 Spacing & Layout

- Bottom navigation bar height: 56dp.
- Side drawer width: 80% of screen width in portrait, fixed 280dp in landscape.
- Task card padding: 12dp horizontal, 10dp vertical.
- Calendar time slot height: 60dp per hour (configurable in Phase 2).
- FAB (Floating Action Button): 56dp, positioned bottom-right, color matches current list accent.